

Tianrui (Dennis) Wu

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EDUCATION

University of Arizona

Bachelor of Science in Physics and Mathematics

Tucson, AZ

Jan 2026 – May 2027 (Expected)

- **GPA:** 3.87/4.0, Dean's List, **Global Wildcat Scholarship** (\$10k/year merit award for academic achievement)
- **Coursework:** Stochastic Calculus, Statistical Machine Learning, Advanced Mathematics Modeling, Computational Physics

Duke University

Bachelor of Science in Physics and Mathematics, Computer Science Minor – transferred, not conferred

Durham, NC

Aug 2021 – Dec 2024

- **GPA:** 3.92/4.0, Dean's List with distinction, **Faculty Scholars Finalist 24'** (highest honor, sole physics nominee)
- **Selected Coursework:** Real Analysis, Advanced Probability, Stat. Learning & Inference, Graduate Machine Learning, Game Theory
- **Extracurriculars:** **Math TA** (2022-2024); **Founder & President**, Duke Astronomy Club (2021-2024); **Chapter President**, Society of Physics Students (2022-2024); **APS Student Ambassador** (2022-2024); **Senior Editor**, Duke Vertices Science Journal (2021-2024)

HONORS & AWARDS

Competitive:

- National 1st Place, *US Physicists' Tournament 2026*
- Top Gold, *British Physics Olympiad 2021 (top 10)*
- Third Grand Award, *Regeneron ISEF 2021 (top 10)*

Awarded:

- J. B. Burnell Outstanding Leadership Scholarship 2024 (sole recipient)
- Member 2024, *ΣΠΣ Physics Honor Society* (reserved for top academics)
- Outstanding Chapter Award 2023, *Society of Physics Students*

EXPERIENCE

Origin Capital (stealth start-up prop shop, \$35M)

Remote

Quantitative Researcher, Systematic Trader, Full-Time

Aug 2025 – May 2026, May 2026-Present (Part-time)

- **Alpha & Directional Strategy:** Built an LSTM forecasting engine on phase-space features with full MLOps (MLflow), integrated 200+ multi-frequency factors with decay analysis and cross-sectional ranking, lifting win rates 3.1%; developed HFT taker strategy with tick-data, capturing short reversions with confidence signal, deployed live at post-fee Sharpe 8.2 over 3 months on Coinbase Intl.
- **Execution & Market Making:** Architected execution algorithms with microstructure timing overlays modulating child-order aggressiveness, cutting slippage ~2 bps vs TWAP across major CEXs; built a dual-engine tick-based backtest stack (vectorized sweeps, L2 queue-position fills) to validate a grid market-making strategy at 3-month out-of-sample Sharpe 9.
- **Trading Operations:** Ran CTA books across OKX, Binance, and Coinbase, tuning algorithmic parameters and discretionary risk to market conditions and strategy factors while owning daily operations (P&L monitoring, TCA, fee analysis).
- **Risk Management:** Directed multi-strategy risk across \$35M in prop/client capital, monitoring live risk across leverage with Grafana dashboard, position, and beta limits with pre/post vol and model attribution, trimming exposure ahead of October crash.

Applied Technologies: Python (PyTorch, MLflow, Polars, pandas, NumPy), CEX APIs (REST/WebSocket), Linux, Git

InsightCheck Investments (systematic equity fund with CTA desk, \$300M AUM)

Shenzhen, China

Quantitative Research Intern

Feb 2025 – Aug 2025

- **Statistical Arb:** Enhanced quantitative models across equity/futures markets and developed market-neutral L/S equity strategy combining PCA statistical factors with mean-reversion signals & covariance-based risk model, backtested with post-cost Sharpe of 2.4.
- **Factor Research:** Researched fundamental and microstructure-based alpha factors for Chinese A-shares, validating 62+ intraday factors with daily IC >0.01 and conducting cross-sectional portfolio optimization in Polars.
- **Orderbook Engine & Infra:** Solely engineered high-performance orderbook reconstruction engine using L3 market-by-order data, prototyping in Python and migrating to C++, achieving ~1 second-per-day process time of full A-share universe.

Applied Technologies: Python (Polars), C++, SQL, Linux, Git

PROJECTS

Density-field Forecasting with High Redshift Uncertainty | Python (pandas, NumPy, Numba), Linux HPC

2026

- Led a first-author cosmological research project as sole analyst, coordinating with a PI and postdoctoral researcher; diagnosed and corrected a systematic bias in a Bayesian MCMC inference pipeline via cross-validated covariance estimation across a 250-cell parameter sensitivity grid, cutting the bias 6-fold. First-author paper in prep: *BAO Reconstruction with Photometric Redshift, PRD (2026)*

IMC Prosperity 4 Trading Competition | Python (PyTorch), Rust

2026

- Ranked top 3% of 6,000+ teams using market-making and cross-asset arbitrage driven by fingerprinting of counterparties.

Crypto Market Maker with Backtest Engine | Python (pandas, Polars, NumPy, Numba, PyTorch), Rust

2025

- Built and deployed live ETH market making framework on OKX, combining EWMA flow alphas with CNN capturing orderbook patterns via phase-space feature representation, and, averaged 0.5-1% daily return. *Project led to recruitment at Origin.*

Neural Network Prediction of Power Spectrum Window Function | Python (PyTorch), Linux HPC

2024

- As Caltech SURF Research Fellow with SPHEREx team, built a ResUNet CNN (PyTorch) for signal deconvolution on large-scale data, owning mock-data generation, distributed HPC training, and validation against analytic baselines at sub-percentage error.

Constraining Dark Matter with Weak Lensing Cosmology and Novel Distance Ladder | Python, Linux HPC

2022-2024

- As a research assistant at Duke Cosmology Group, built Python pipelines generating stellar catalogs and analyzing 120 TB of simulations to extract features for distance calibration and weak lensing maps, applying MCMC-based Bayesian inference to constrain params, leading to published papers: *JWST, ApJ (2024)*; *OpenUniverse2024, MNRAS (2025)*

SKILLS & DOMAIN

Languages: Python (PyTorch, Polars, pandas, NumPy, Numba, Dask, MLflow), Rust, C/C++, PostgreSQL, Unix; English, Mandarin

Market Exposure: Digital Assets, Commodity Futures, Global Equities, Options (completed *Akuna 101/201*), Prediction Markets

Interests: Poker, Competitive GeoGuessr, Tabletop Wargaming, Strategy Board Games, Chaos Theory, Forecasting, Astrophotography, Sci-fi